05-ELECTRON SPIN RESONANCE (ESR)

PURPOSE of the EXPERIMENT

This experiment aims to investigate the magnetic moment of the electron spin.

1. Determination of the Landé-g factor of the DPPH (Diphenylpicrylhydrazyl) sample.

2. Determination of the half-width of the absorption line (FWHM).

GENERAL INFORMATION

Electron spin resonance (ESR) is based on the energy absorption by substances with unpaired electrons in an external magnetic field generated by a DC source.

Determination of Landé-g Factor

There are two different atomic magnetic dipoles. Circular currents represent the circulation of electrons around the atomic nucleus, and magnetic moments depend on the electron spin.

The magnetic moment generated by a circular current,

$$\vec{A} = A.\vec{e} = \pi.R^2.\vec{e}$$

Here A is the surface of the region where the electron moves,

R, orbital radius,

$\vec{e}$ is the unit vector perpendicular to the surface.

ω, is the angular velocity, $T = 2\pi / \omega$ is the period of the orbit, and the orbital magnetic moment is

$$\vec{\mu}_{orbital} = I.\vec{A} = -\frac{e}{T}.\pi.R^2.\vec{e} = -\frac{e.\omega}{2}.R^2.\vec{e} \qquad (1)$$

given by the equation.

The magnetic moment is directly proportional to the angular momentum $\vec{L}$. m_e is the mass of the electron;

$$\vec{L} = m_e.\vec{r}x\vec{v} = m_e.\omega.R^2.\vec{e} \qquad (2)$$

given by the equation.

If this expression is substituted in equation (1),

$$\vec{\mu}_{orbital} = -\frac{e}{2.m_e}.g_{orbital}.\vec{L} = \gamma_{orbital}.g_{orbital}.\vec{L} \qquad (3)$$

is obtained. Here, $g_{orbital}$ is the "Landé Factor" and $\gamma_{orbital} = -\frac{e}{2.m_e}$ is called the "Gyromagnetic Ratio". A comparison of equations (1) and (2) shows that the momentum of a pure orbital Landé-factor $g_{orbital}$ = 1 is obtained.

$$\vec{\mu}_{orbital} = \gamma_{orbital}.\vec{L} \qquad (4)$$

The spin angular momentum $\vec{S}$ represents the spin of an electron. The electron's magnetic moment is directly proportional to this spin angular moment.

$$\vec{\mu}_{spin} = -\frac{e}{2.m_e}.g_{spin}.\vec{S} = \gamma_{spin}.\vec{S} \qquad (5)$$

$$\gamma_{spin} = -g_{spin}.\frac{e}{2.m_e} = g_{spin}.\gamma_{orbital}$$

One purpose of this experiment is to obtain the Landé-g coefficient for the electron spin. Just as electrons have spins, nuclei also have spins. However since the mass of the nucleus is much larger than the mass of the electrons, $\gamma_{çekirdek}$ is very small. Therefore the magnetic moment of the nucleus can be neglected.

According to quantum mechanics, angular momentum takes discrete values.

$$L_Z = m\hbar \qquad (m = l, l\text{-}1, l\text{-}2, \ldots..1\text{-}l, l) \qquad (6)$$

Where *m* is the magnetic quantum number, h=6.63.10^{-34} Js is Planck's constant and $\hbar \equiv \frac{h}{2\pi}$.

The numerical value of the spin angular momentum can only be 1/2.

$$S_Z = \pm\frac{1}{2}\hbar \qquad (7)$$

Magnetic moments are also quantized and defined in terms of the Bohr magneton.

$$\mu_B = \frac{e}{2.m_e}.\hbar = 9.27x10^{-24}\,Am^2 \qquad (8)$$

z-components of orbital and spin angular momenta are,

$$\mu_{z,orbital} = \gamma_{orbital}.L_z = -m.\mu_B \qquad (m=l, l\text{-}1, l\text{-}2, \ldots.1\text{-}l, l) \qquad (9)$$

$$\mu_{z,spin} = \gamma_{spin}.S_z = g_{spin}.\gamma_{orbital}.S_Z = \left(\mp\frac{1}{2}\right).g_{spin}.\mu_B \qquad (10)$$

given by these equations.

Total angular momentum if an electron has both orbital angular momentum and spin angular momentum,

$$\vec{J} = \vec{L} + \vec{S} \text{ ve } |J| = |L-S|, |L-S+1|, |L-S+2|, ..., |L+S|$$

The corresponding total magnetic moment,

$$\vec{\mu}_j = \vec{\mu}_l + \vec{\mu}_s = -\frac{e}{2.m_e}.\vec{L} - \frac{e}{2.m_e}.g_{spin}.\vec{S} = -\frac{\mu_B}{\hbar}.\left(\vec{L} + g_{spin}.\vec{S}\right) \qquad (11)$$

If L and S values are known, the Landé-g coefficient can be calculated as follows:

$$g_j = 1 + \frac{J.(J+1) + S.(S+1) - L.(L+1)}{2.J.(J+1)} \qquad (12)$$

The DPPH sample used in the experiment has one unpaired electron. Its orbital magnetic moment $L = 0$ and the total magnetic moment consists only of the spin magnetic moment and is $g_{spin} = g_j$. This value is approximately equal to the Landé factor of a free electron. Substituting $S = 1/2$ and $L = 0$ in equation (12) gives $g_j = 2$. However, the actual value of g is slightly larger than 2 due to other interactions that depend on B.

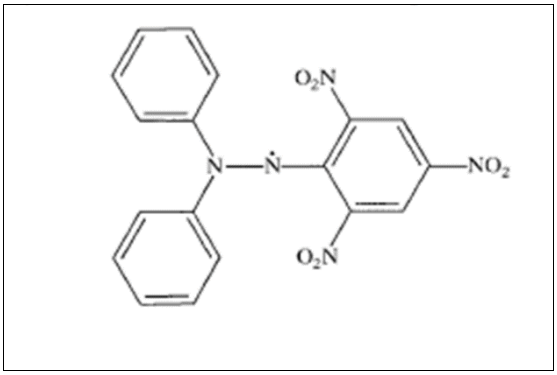


Figure 1: DPPH (Diphenylpicrylhydrazyl) sample

Zeeman Effect and Magnetic Resonance

The Zeeman Effect describes the splitting of the atomic spectral line due to an external magnetic field applied to an atom or molecule. This effect is caused by the interaction between the

magnetic moments of the atom or molecule and the applied external magnetic field. The following expression applies to the potential energy in a magnetic field:

$$E = -\mu.B \tag{13}$$

There are two types of Zeeman effects, Normal Zeeman ($S = 0$) and Anomalous Zeeman ($S \neq 0$). Since only the orbital angular momentum (L) exists in the normal Zeeman effect, only the orbital magnetic moment and the external magnetic field interact. When the external magnetic field is applied, the energy levels of the atom split into equally spaced (ΔE) levels:

$$\begin{aligned} \Delta E &= -\Delta\mu_{orbital}.B \\ &= \frac{e}{2.m_e}\Delta L_z.B \\ &= \mu_B.\Delta m.B \end{aligned} \tag{14}$$

But as in this experiment, the electron spin must also be taken into account. Thus, the magnetic field interacts not only with the orbital magnetic moment but also with the spin magnetic moment. In this case, the Zeeman interaction is written in the following form:

$$\Delta E = \mu_B.\Delta m_j.g_j.B \tag{15}$$

With the selection rule $\Delta m_j = \mp 1$ for magnetic transitions, the distance between two Zeeman levels is as follows:

$$\Delta E = \mu_B.g_j.B \tag{16}$$

For the transition from a low energy level to a high energy level to take place, energy must be absorbed whose absolute value is equal to the energy difference between the two levels. This energy comes from the electromagnetic wave of frequency f provided by the ESR resonator. This energy,

$$E = hf \tag{17}$$

This process is called “Magnetic Resonance”.

The resonance condition is achieved by varying the B_r magnetic field by changing the voltage applied to the Helmholtz coils during the experiment.

The Landé factor is calculated according to the following equation:

$$\mu_B.g_j.B_r = hf$$

$$g_j = \frac{h.f}{\mu_B.B_r} \tag{18}$$

Here; h = 6.63.10^{-34} Js, f = 146 MHz, μ_B = 9.27.10^{-24} Am^2, if these values are substituted;

$$g_j = 10{,}43.10^{-3} T.\frac{1}{B_r} \tag{19}$$

is obtained. Here T is the abbreviation for tesla.

The magnetic field in the ESR experimental set used is provided by a pair of Helmholtz coils of turns ω = 241 and radius R = 4.8 cm.

$$B = \mu_o.\frac{8.I.\omega}{\sqrt{125.R}} = 0.7155.\mu_o.\frac{I.\omega}{R} \tag{20}$$

Here μ_0 = 4π10^{-7} Tm/A and I is the current through the coil. If the values are substituted;

$$B = 0.6445.\mu_o.\frac{I.\omega}{R}$$

$$B = 4.07.10^{-3} I \ \frac{T}{A} \tag{21}$$

is obtained. The value of g is calculated by substituting the resonant current I_r into equation (22):

$$g = \frac{2.565A}{I_r} \tag{22}$$

HOW TO CONDUCT THE EXPERIMENT

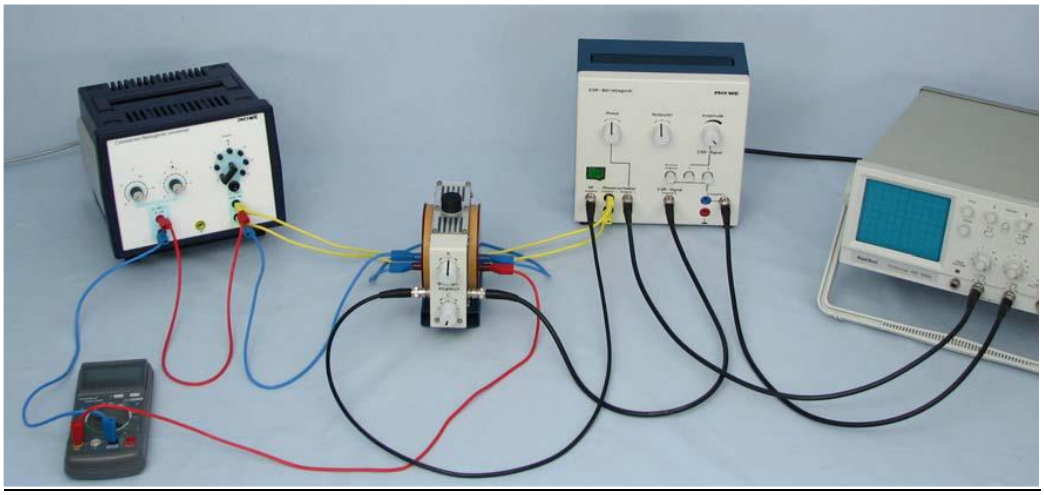


Figure 2: ESR experimental setup

Determination of Landé-g Factor

The ESR test setup is shown in Figure 2.

1. Make sure that the direct voltage (DC) switch labeled "V" on the main power unit is set to "0" and the current switch labeled "A" is set to "5A".

2. The alternating voltage must be set to 2V (this corresponds to a frequency of 50 Hz).

3. It is important that the direct voltage is overlapped by the alternating voltage so that the ESR signal is displayed on the oscilloscope.

4. Then switch on the main power unit, the ESR power unit and the oscilloscope.

5. Press the "Bridge Balancing" button (button 8) on the ESR power unit (Figure 3).

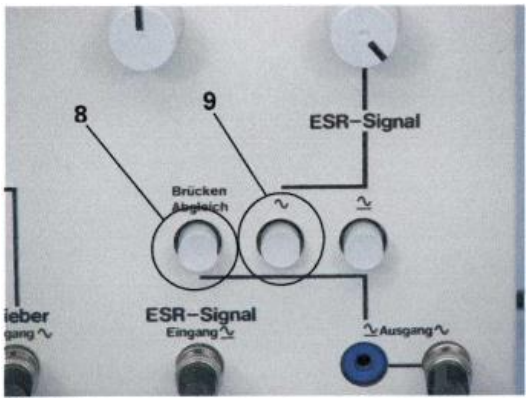


Figure 3: Part of the ESR power unit

6. The "R" switch of the resonator should be in the center position and the "C" switch in the far left position.

7. On the oscilloscope, select the "X-Y" mode (Figure 4).



Figure 4: "X-Y" mode

8. Select "GND" mode for channel "X" and "d.c." mode for channel "Y".

9. The signal sensitivity for both channels should be 1 V/cm (Figure 5).



Figure 5: The signal sensitivity

10. A single point should now be visible on the oscilloscope.

11. Move the "Position" switch to the point displayed in the center of the coordinate system.

12. Next, press button "9" on the ESR power unit (Figure 3) and select the d.c. mode for channel "X" on the oscilloscope.

13. You should now see a horizontal line on the oscilloscope.

14. Increase the direct voltage on the main power unit until the digital multimeter reads 1.3 A.

15. Now carefully turn the "C" knob on the resonator to the right until a signal appears on the oscilloscope. (It may be important to increase the intensities of the X and Y channels to 0.5 V/cm or more to obtain a stronger signal.)

16. As soon as the signal appears, the two lines coincide with the "Phase" button of the ESR power unit.

17. Adjust the signal with "C" until you see a symmetrical shape with a minimum. (Try to make the signal as symmetrical as possible.)

18. The minimum point in the figure should be brought exactly to the y-axis by reducing the direct voltage on the main power unit. (Use "C" again to get a symmetrical figure.)

19. If the figure is as follows (Figure 6), you have obtained a good "resonant signal".

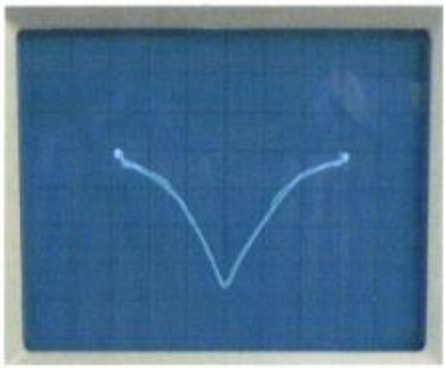


Figure 6: The resonant signal

20. The current value read from the multimeter at the moment when this resonant signal is obtained is called "Resonant Current" (I_r).

21. Record this I_r current in Table 1. Repeat all the steps so far two more times and record the I_r currents you have found in Table 1.

22. Substitute these current values in the equation $g = \frac{2.565A}{I_r}$ and calculate the value of g with error and record it in Table 1. The value in the literature is g = 2.0037.

Table 1: Results

I_{r1}	g_1	I_{r2}	g_2	I_{r3}	g_3
			g_{ave} =	±	

Determination of FWHM (Full Width at Half Maximum)

1. Use the "Position" switches to adjust the position of the signal so that the x-axis passes exactly half the height of the signal.

2. Starting from the zero point, count the scaled divisions until the signal and x-axis coincide. The more symmetrical the signal, the more precise and accurate this measurement will be).

3. The number of these scaled divisions depends on the sensitivity you have chosen. Therefore, make sure that you do not change the signal sensitivity during the rest of the experiment.

4. Record your results below.

5. Intersection (RIGHT SIDE) =

6. Intersection (LEFT SIDE) =

7. To determine the half-width (FWHM), the distance between the two intersections must be measured in Amperes.

8. Therefore, the alternating voltage must be switched off and the resonator must be connected directly to the direct voltage. For this purpose, the red wire connected to the direct voltage input with the alternating voltage input must be disconnected and the blue wire of the resonator must be connected to the now empty direct voltage input (Figure 7).

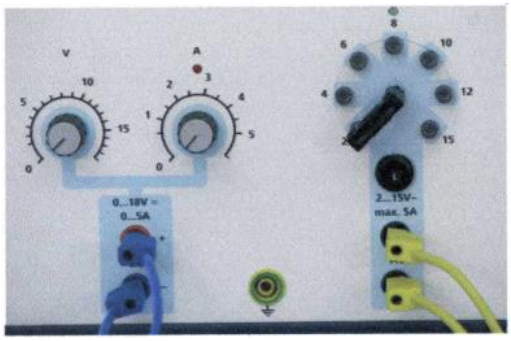


Figure 7: Part of the power supply

9. Disconnect the BNC cable from the ESR power unit connected to the "X" channel of the oscilloscope.

10. Connect the appropriate adapter to the end of this BNC cable as shown in Figure 8.



Figure 8: BNC cable

11. Then connect this adapter to the direct voltage inputs of the main power unit (Figure 9).



Figure 9: The direct voltage inputs

12. Make sure that the sensitivity of the "X" and "Y" channels is not changed during this experiment.

13. Vary the correct voltage until only a single dot appears on the oscilloscope.

14. When the dot appears, use the "position" knob to change its position until the dot is on the x-axis.

15. Move the point you have found to one of the two intersections you found earlier by changing the correct voltage on the main power unit.

16. Note the current value (I_1) read from the multimeter at this moment.

$$I_1 =$$

17. Now move the point to the other intersection and note the second current (I_2).

$$I_2 =$$

Calculate the absolute value of the difference between the currents I_1 and I_2 to obtain the width at half maximum. Write down your results below.

$$\Delta I = \left| I_1 - I_2 \right|$$

18. Then calculate the magnetic displacement by writing the value of ΔI in equation (21) and note your result below. This value you found is the FWHM. The value in the literature is $\Delta B = 2{,}8.10^{-4}$ Tesla.

$$\Delta B = \ldots\ldots\ldots\ldots$$

REFERENCES

1. http://www.nikhef.nl/~h73/kn1c/praktikum/phywe/LEP/Experim/5_1_12.pdf
2. Phywe experimental literature and operating manuals